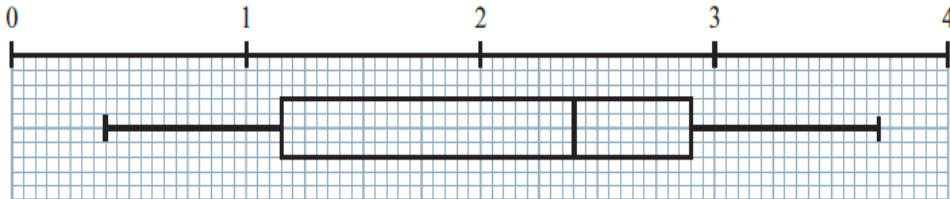




United International University
School of Science and Engineering
Department of Computer Science and Engineering
Mid Term Examination Trimester: Spring – 2025
Course: Math -2205 (Probability & Statistics)
Total marks – 30 || Duration – 1 hour 30 minutes

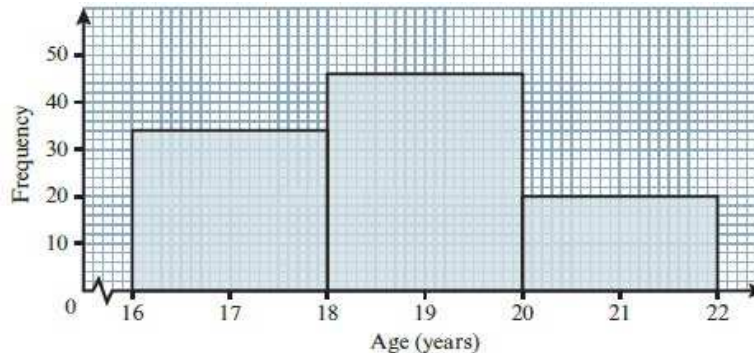
[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

- Q1 (a)** (i) Find the range and the interquartile range of the dataset represented in the following box plot [2]



- (ii) What type of skewness would you expect this set of data to have. [1]

(b)



Construct the corresponding group frequency distribution for the above histogram. Hence find the mode and the arithmetic mean of the distribution. [5]

- (c) If the second and fourth Central moment of a distribution is 3 and 27 respectively, find the **Kurtosis** and hence state the shape of the distribution. [2]

- Q2 (a)** Discuss the strength of correlation from the following Pearson' correlation coefficient. (i) $r = 0.35$ (ii) $r = -1$ [2]

(b) A company wants to analyze the relationship between the number of hours studied (x) and the score obtained in a test (y) by 6 students. The data collected are as follows:

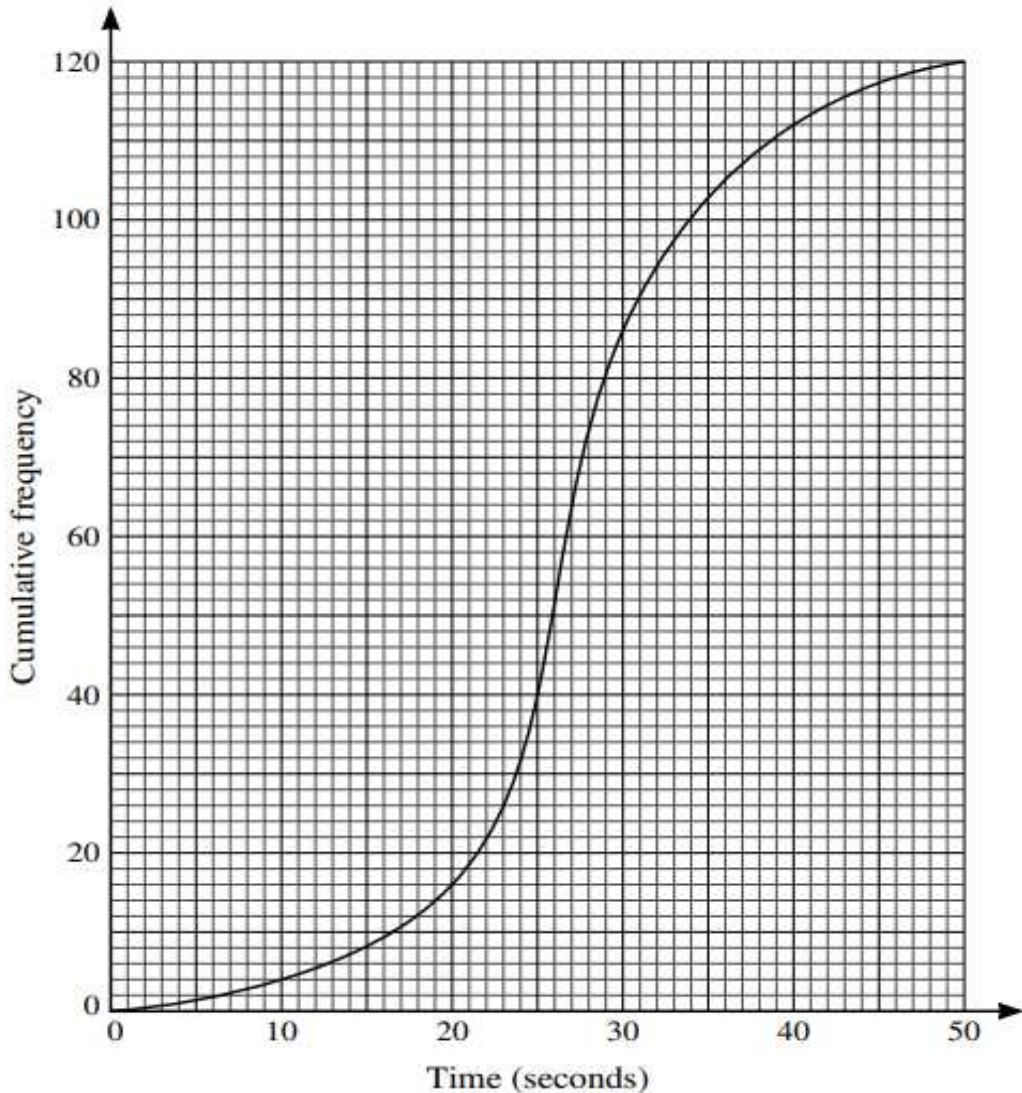
Hour Studies (x)	2	3	5	8	9	10
Test Score (y)	65	70	75	85	95	98

- (i) Find the value of the coefficient correlation (r) and interpret the result. [3]
(ii) Using the method of least squares, find the equation of the linear regression line in the form: $y = a + bx$ [4]
(iii) Estimate the expected score of a student who studies for 6 hours using the regression model. [1]

- Q3** For the dataset denoted by x in the following table (where k , is a constant) it is given that the arithmetic mean of x i.e. $\bar{x} = 9.85$. Find the value of k , and hence calculate the variance of x . **[5]**
- (a)** .

x	5	7	8	12	20
f	k	$2k$	$k+1$	4	3

- (b)** The times taken by 120 children to complete a particular puzzle are represented in the following cumulative frequency graph.



- (i)** Use the graph to estimate the median and 70th percentile. **[2]**
- (ii)** Find the interquartile range **[2]**
- (iii)** 20% of the children took less than T seconds to complete the puzzle. Use the graph to estimate the value of T. **[1]**

Formulae:	For the regression line: $y = a + bx$	$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$ and $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$
	Coefficient of correlation	$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$
	Spearman's rank correlation coefficient:	$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$